

Abhinav

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Education

Kalinga Institute Of Industrial Technology	Bhubaneswar, OD , IN
BTech in Computer Science and System Engineering	08/2024 - 06/2028

Projects

Container Provider (TempDev)	Go, Docker, WebSockets, Cloudflared	04/2026 - 05/2026
◦ Built a cloud sandbox engine spinning up fully isolated Linux environments in the browser in under 3 seconds ◦ Engineered dynamic Cloudflare tunnel provisioning to instantly expose internal container ports to public HTTPS URLs ◦ Designed a database-free, in-memory scheduler using Go channels, running efficiently on AWS Free Tier ◦ Hardened security by dropping kernel capabilities (<code>--cap-drop=ALL</code>) and building an automated abuse sweeper		
Rta Code Editor	JavaScript, Python, FastAPI, Electron	04/2026 - Present
◦ Architected an AI-assisted terminal agent maintaining an ultra-lightweight memory footprint (under 20MB) ◦ Engineered a local Python middleware proxy to seamlessly aggregate commercial and local LLM APIs ◦ Designed context-compilation strategies to efficiently feed large code snippets into AI engines		
PMG (Poor Man's Git)	Go, FastAPI, React	10/2025 - 12/2025
◦ Built a custom Version Control System CLI from scratch in Go to track and version local repository directories ◦ Developed an indexing engine utilizing SHA256 mapping to track delta changes, similar to Git's object database ◦ Created a React web dashboard connected via FastAPI to visually map commit histories and branch topologies		
Elin Programming Language	Python, C++, Virtual Machine	01/2026 - Present
◦ Constructed a Python parser and compiler to translate source code into a custom stack-based bytecode ◦ Engineered a fast, low-overhead C++ Virtual Machine to natively execute the custom Instruction Set Architecture ◦ Implemented arithmetic operations, loops, and stack routines, showcasing core compiler design principles		
Microcontroller OS	C++, ESP8266, LittleFS	09/2025 - Present
◦ Designed an experimental firmware shell for NodeMCU boards operating directly over physical serial connections ◦ Integrated a custom wrapper for LittleFS, achieving high-performance flash memory indexing and wear-leveling ◦ Exposed serial CLI commands for file system manipulation, real-time data transfers, and hardware pin controls		

Skills

Programming Languages:  Go,  Python,  C++,  JavaScript ,  C , Odin

Tools and Frameworks: Docker, FastAPI, React, Electron, WebSockets, Linux (PTY, cgroups), Cloudflared, LittleFS, Git

Languages: English (Proficient) , Hindi (Native)